Arrays

15-3-26

Aim – Demonstrate Arrays(list) in python.

Theory – Arrays in Python are commonly implemented using lists to store multiple values in a single variable. Lists allow accessing elements using indexes, modifying values, adding new elements with methods like append(), and removing elements. They are dynamic, ordered, and support iteration, making them useful for storing and managing collections of data.

A1.

```
# Creating an array (list) of integers

numbers = [10, 20, 30, 40, 50]


# Accessing elements

print("First element:", numbers[0])

print("Last element:", numbers[-1])


# Modifying an element

numbers[2] = 35

print("Modified array:", numbers)


# Adding elements

numbers.append(60)

print("After appending 60:", numbers)


# Removing elements

numbers.remove(20)

print("After removing 20:", numbers)


# Iterating through the array

print("All elements:")

for num in numbers:

    print(num)
```

```
First element: 10
Last element: 50
Modified array: [10, 20, 35, 40, 50]
After appending 60: [10, 20, 35, 40, 50, 60]
After removing 20: [10, 35, 40, 50, 60]
All elements:
10
35
40
50
60
```

Conclusion

Python lists provide a flexible way to store and manipulate multiple values, allowing easy access, updates, additions, deletions, and iteration within programs.